Jacob Stephens

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Software and infrastructure engineer building software factories and reliable AI-agent workflows. Built thirdshift, an open-source Rust orchestrator for unattended issue-to-PR delivery. Lead a two-engineer team responsible for a specialty-travel company's reservations, payments, Linux fleet, and agent infrastructure, with 4+ years of production ownership.

Current Focus: Software Factories

thirdshift | Open-source Rust orchestration

Source & architecture | Review evaluation: 183 sessions

- Built a deterministic orchestration engine across six coding-agent harnesses, coordinating implementation, standards and spec reviews, CI checks, and bounded repair loops; merging is explicitly configured.
- Implemented dependency-aware parallel ticket execution in separate Git worktrees, resumable runs, and architecture-improvement and security-audit workflows. Failed runs preserve work for continuation.
- Designed an evaluation of 183 replayed code-review sessions with fresh-session controls and predefined acceptance bars. Published results and limitations to guide where to add a second review, rather than assuming a different model family improves reliability.

Experience

Head of Engineering

Educational Travel Adventures | Feb 2022 - Present | Remote

- Lead a two-engineer team with end-to-end responsibility for architecture, operations, deployment, and on-call response across the production stack of a multi-million-dollar travel business.
- Built three Docker/Traefik Manager Sandboxes with separate application containers, a shared writable database refreshed daily, and per-manager credentials. Shipped 14 manager-prototyped features through human review.
- Built an operations console (Python, Flask, SQLite) for fleet health, deploy review, branch previews, and browser terminals, with Google sign-in, passkeys, and per-permission access. Deployed Prometheus, Grafana, and Alertmanager across 14 hosts.
- Reduced group-manifest SQL statements per request from about 2,650 to 183 with byte-identical output. A separate measurement-led index pass eliminated about 80% of measured query time on the live primary using online DDL.
- Shipped a mobile-first guide PWA with offline support and receipt-attached expenses: 77 guides filed 410 tour reports and 1,800+ expense reports (about $154K) over a 12-month period.
- Led PHP 5-to-8, MySQL 5-to-8, and CentOS-to-Rocky Linux migrations. Modernized payment integrations and ACH/NACHA workflows, including bank-account encryption, as part of a PCI compliance initiative.
- Built a deployment pipeline with dirty-worktree rejection, serialized deploys, ancestor-only refs, health checks, and rollback; validated with PHPUnit and GitHub Actions across PHP 8.1-8.4.

Web Developer and Client Support Specialist
Sharp Innovations | Nov 2021 - Feb 2022 | Conestoga, PA

- Automated SQL/PHP database migrations, reducing execution from hours to minutes; supported client websites, security fixes, and Linux server configuration.

Full-Stack Developer (Independent)
Steward Goods | Mar 2020 - Nov 2021 | Remote

- Designed and shipped a multi-tenant possessions tracker in PHP and MySQL with full frontend, backend, and deployment ownership; implemented Stripe subscription payments for client work.

Selected Projects

- vaulted-agent: Launches coding agents with vault-resolved secrets injected in-process, per-agent manifests, and vault-manager tokens removed from the child environment. Supports 1Password, Bitwarden Secrets Manager, pass, and SOPS.
- infrastructure-patterns: Public, sanitized architecture decisions covering agent isolation, scoped credentials, human merge gates, audit trails, and rollback.
- Infrastructure as code: Imported about 220 live DNS records across nine zones into Terraform without recreating them; added S3 remote state, Ansible provisioning, and plan-on-PR checks.
- Cascade: A headless Rust audio core shared by four released shells (web PWA, Android, Windows, macOS) through a single JSON interface, with wasm-bindgen, UniFFI, and C ABI bindings; iOS and watchOS shells are in TestFlight.
- Chart35: A shipped fertility tracker with client-side AES-256-GCM encryption, offline-first storage, and expiring provider-share links.

Core Technologies

LANGUAGES & DATABASES
Rust, Python, TypeScript/JavaScript, PHP, SQL, MySQL, PostgreSQL, SQLite

AGENT & DELIVERY SYSTEMS
Coding-agent orchestration, Git worktrees, review evaluations, CI/CD, GitHub Actions, PHPUnit, Vitest, Playwright

INFRASTRUCTURE & APPLICATIONS
Linux, Docker/Compose, Traefik, Terraform, Ansible, Prometheus/Grafana, FastAPI, Flask, React/Next.js, REST APIs

Education & Certification

BA, Sociology | Gordon College | 2017
CompTIA A+ | Earned November 2017